Grade 6 Accelerated Name_Answer Key________
Unit 11 Date___________________
Lesson 6 Homework Period__________________

1. Sophia purchased a new car for $\$26{,}714.00$. She read that the value of new cars change by $\frac{-4}{19}$ in the first year. What is the value of Sophia's car after the first year?

$26{,}714\left(\frac{15}{19}\right) = \$21{,}090$ is the value after the first year.

2. Sophia learns that the value of her car will continue to change by $\frac{-3}{17}$ of its value each year after the first year. What is the value of Sophia's car at the end of year 2?

$21{,}090\left(\frac{14}{17}\right) = \$17{,}368.24$ is the value at the end of the second year.

3. An Orca whale is swimming $-250\frac{3}{4}$ ft below sea level when he notices the rest of his pod above him at the surface. Orca whales can swim at a rate of 183.74 feet per hour. How long, in hours, will it take the Orca to reach his pod?

$\frac{183.74 \text{ ft}}{1 \text{ hr}} = \frac{-250\frac{3}{4} \text{ ft}}{x \text{ hr}}$ $x \approx 1.4$ hours

4. A candle store sells the 3-wick candles for $3\frac{1}{3}$ times what they paid for them from the warehouse. The candles are purchased from the warehouse at $\$6.75$. How much profit does the candle store make off each candle?

$6.75 \times \frac{10}{3} = \$22.50$ $22.50 - 6.75 = \$15.75$ profit

5. After 6 months, the price of the candles is reduced by $\frac{1}{5}$ of the original store price. What is the price of the candles after the 6 month reduction?

$$22.50 - \left(\frac{1}{5} \cdot 22.50\right) = \$18.00$$

6. Priscilla is planning to hike a new trail this weekend. She sets up camp Friday evening at the start of the trail which begins at an elevation of -184.56 ft and ends at an elevation of 759.22 ft. She plans to begin her hike in the early morning and wants to travel $\frac{5}{8}$ of the change in elevation of her trip on Saturday, then set up camp until the next morning. What is Priscilla's starting elevation on Sunday?

$759.22 - (-184.56) = 943.78$
$\frac{5}{8}(943.78) = 589.86$ ft. on Sat.

$-184.56 + 589.86 = 405.3$ ft is the starting elevation on Sunday.

7. Kate is running a race on Monday, which has a total distance of 26.13 miles. Kate's family wants to show their support by holding up encouraging signs along the path. Her mother decides to stand at the spot that is $\frac{4}{13}$ of the total race while her father chooses to stand at the spot that is $\frac{10}{13}$ of the total race. How many miles apart are Kate's parents standing from each other?

$26.13\left(\frac{4}{13}\right) = 8.04$ miles

$20.1 - 8.04 = 12.06$ miles apart

$26.13\left(\frac{10}{13}\right) = 20.1$ miles

8. Kate runs at a steady rate of 12.3 minutes per mile. Using the information given in Question 7, how many minutes after the race starts should her mother expect to see Kate run by?

$\frac{8.04 \text{ miles}}{x \text{ min}} = \frac{1 \text{ mile}}{12.3 \text{ min}}$ $\quad x = 12.3(8.04) = 98.892$ minutes

9. Joshua used his store credit card to purchase a new basketball. He has a coupon that when he uses his credit card, the item will be $\frac{-1}{3}$ of the total. The basketball was priced at $\$21.39$. A month goes by and Joshua forgot to pay his credit card bill, so they have added a fee of 0.29 times the amount he owed. What is the total amount Joshua has to pay his credit card company?

$21.39\left(\frac{2}{3}\right) = 14.26$
$14.26 \cdot 1.29 = 18.40$

Joshua owes $\$18.40$

10. Sandy is moving to Jacksonville, Florida from New York, New York. The total distance of her move is 945.61 miles. She rented a moving truck that charges $\$19.95$ a day and $\$0.35$ per mile over 500 miles. If she returns the truck with a full tank of gas, the company will discount her bill by $\frac{-2}{31}$. What is the total amount Sandy will pay for the moving truck if she fills the tank with gas before returning it?

Assuming she drives it all in one day:
$19.95 + 0.35(945.61 - 500) = 175.91$
$175.91\left(\frac{29}{31}\right) = \$164.56$ is Sandy's total bill for the moving truck.